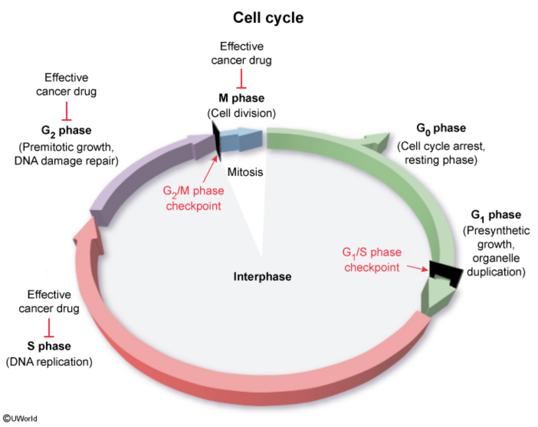
A subset of aggressive cancers has a relatively high growth rate, leading to the formation of large tumors. An effective drug against fast-growing tumors would most likely NOT target which stage of the cell cycle?

- A. G_0 [83%]
- B. G_2 [6%]
- C. G_2/M checkpoint [6%]
- D. G_1/S checkpoint [4%]

Correct
83% answered correctly

Explanation:



The mechanism by which cells duplicate their content and divide is **controlled** by "**checkpoints**" (restriction points) within the cell cycle.

Many **cancers** arise from the accumulation of multiple genetic defects (eg, inherited or from mutations) that allow **uncontrollable cell division** without the normal rate of apoptosis (programmed cell death). According to the question, a subset of aggressive cancers can cause the growth of large tumors, masses of abnormal cells. Because the cells within these tumors are actively dividing, it can be assumed that the majority of the cells are in interphase and **not arrested in G_0**. Therefore, to treat these aggressive cancers, clinicians may use anticancer drugs that try to destroy **actively dividing cells (phases G_1 to M)** and not those in **G_0**.

(Choice B) Cell growth, protein synthesis, and DNA damage repair occur in preparation for mitosis during the G_2 phase of actively dividing cells. Cancer cells divide at higher rates than normal cells and cycle through G_2 frequently. Therefore, anticancer drugs that target G_2 can decrease the number of cell divisions. As a result, survivability may be improved because cancer cells halted at G_2 are less likely to disrupt physiological function and metastasize (spread).

(Choices C and D) Control of cellular division occurs at checkpoints within the cell cycle. At the G_1/S phase transition, the cell commits to undergoing a division cycle whereas the G_2/M phase transition acts as a quality-control step in which the cell cycle is stopped and cellular components are checked for abnormalities. Checkpoints are regulated primarily by cyclins and cyclin-dependent kinases. Cells with abnormalities that cannot be repaired undergo apoptosis; those that pass the checkpoints continue to divide. Anticancer drugs often target these checkpoints due to the regulatory roles on the cell cycle.

Educational objective:
Most cells in the human body are arrested in G_0. However, cellular transition into G_1 prepares a cell for division and DNA synthesis (S phase). In the G_2 phase, DNA is checked for errors and the cell ensures that sufficient organelles and cytoplasm are available for cell division. Subsequently, the cell divides in the M phase via mitosis and cytokinesis. Compounds that inhibit cell division typically target the cell cycle in phases G_1 to M.

Time spent: 0 sec
Subject: Biology

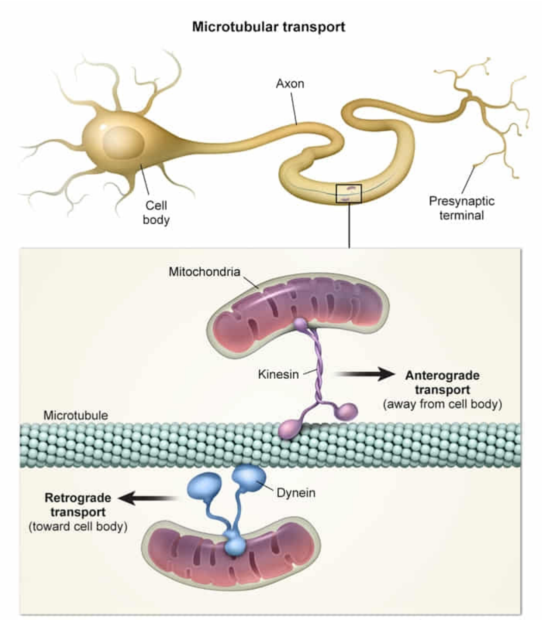
QID:401266
System: Reproduction

In a neuron, mitochondrial biogenesis is believed to occur primarily in the cell body, but mitochondria are often positioned at the presynaptic terminal, a distal site with high metabolic demand. Given this information, which molecular mechanism is most likely responsible for mitochondrial transport from the cell body to the presynaptic terminal?

- A. Kinesin motors transport mitochondria along microtubules [55%] ✔ (selected)
- B. Kinesin motors transport mitochondria along microfilaments [18%]
- C. Dynein motors transport mitochondria along microfilaments [7%]
- D. Dynein motors transport mitochondria along microtubules [18%]

Correct
54% answered correctly

Explanation:



A eukaryotic cell's cytoskeleton is an intracellular scaffolding (network) of fibers interspersed throughout the cytoplasm. The cytoskeleton is composed of three types of fibers: Microfilaments, intermediate filaments, and microtubules. Together, these fibers function to organize cellular components, support cellular motility (eg, cell movement, intracellular transport), and give the cell its shape.

Microtubules are structural cytoplasmic filaments composed of tubulin subunits. These filaments have a number of critical functions, including serving as tracks for intracellular transport of organelles and vesicles. The movement of intracellular cargo along microtubules is mediated by two motor proteins:

- **Kinesin:** Moves intracellular cargo along microtubules in **anterograde** axonal transport (ie, away from the nucleus and toward distal sites).
- **Dynein:** Participates in **retrograde** axonal transport of intracellular cargo (ie, from distal sites toward the nucleus) **(Choice D)**.

According to the question, mitochondria are transported from the cell body **toward** the

presynaptic terminal. Because mitochondria are being transferred from the nucleus and toward the synaptic terminal, this anterograde transport is most likely being performed by **kinesin** motors moving along the **microtubules**.

(Choices B and C) Movement along actin microfilaments occur via *myosin motors* (not kinesin or dynein). The primary functions of myosin movement along microfilaments are cellular locomotion, cellular division, and muscle contractions.

Educational objective:
The intracellular scaffolding of a eukaryotic cell is composed of three families of protein filaments: Microfilaments, intermediate filaments, and microtubules. Intracellular transport of cargo (eg, organelles, vesicles) is mediated primarily by two microtubular motor proteins (kinesin and dynein). Kinesin mediates anterograde transport (ie, away from the nucleus) whereas dynein mediates retrograde transport (ie, toward the nucleus).

Time spent:0 sec	**QID**:401267
Subject:	**System:**
Biology	Reproduction